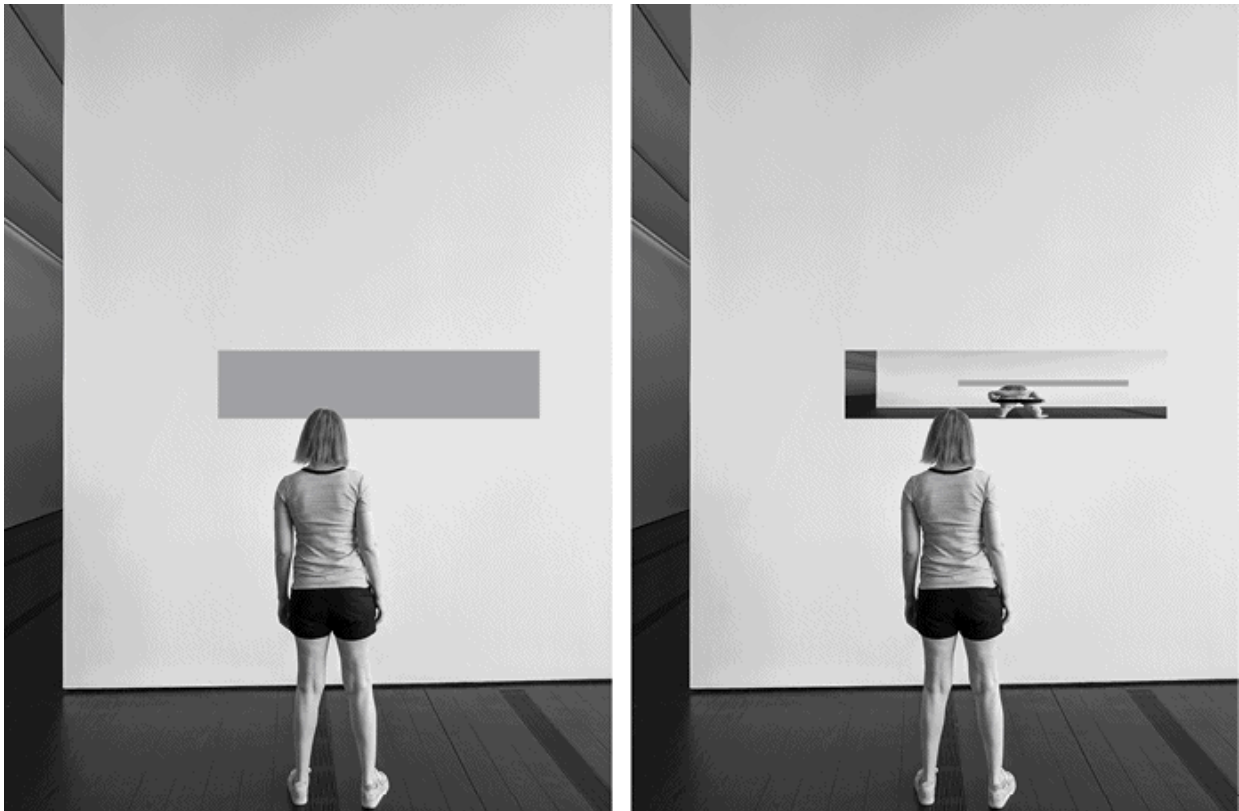


We cannot simply resize the base image to the dimensions of the magenta area because it's unlikely the two share the same *aspect ratio*, or proportion of the width divided by the height. Doing so results in a recursive image that looks stretched or squished, like [Figure 14-5](#).

Instead, we must make the resized image large enough to completely cover the magenta area but still retain the image's original aspect ratio. This means either setting the width of the resized image to the width of the magenta area such that the height of the resized image is equal to or larger than the height of the magenta area, or setting the height of the resized image to the height of the magenta area such that the width of the resized image is equal to or larger than the width of the magenta area.



[Figure 14-5](#): Resizing the image to the dimensions of the magenta area can result in a different aspect ratio, causing it to look stretched or squished.

To calculate the correct resizing dimensions, the program needs to determine the aspect ratio of both the base image and the magenta area:

Python